

● RAG Learning

1.2 Fractions, Decimals & Percentages

Pearson Edexcel GCSE Mathematics 1MA1 — Number — spec refs N2 (fractions), N8 (exact fractions), N10–N12 — Papers 1 (non-calculator), 2 and 3 (calculator)

A fraction, a decimal and a percentage can all describe the same quantity. In 1MA1 you must apply the four operations to fractions and mixed numbers (positive and negative), calculate **exactly** with fractions, switch between terminating decimals and fractions, use fractions and percentages as operators, and handle fractions inside ratio problems. Converting between **recurring decimals** and fractions is **Higher only**.

Quick Reference: Equivalences to Learn

Fraction	Decimal	Percentage	Fraction	Decimal	Percentage
$\frac{1}{2}$	0.5	50%	$\frac{1}{8}$	0.125	12.5%
$\frac{1}{4}$	0.25	25%	$\frac{5}{8}$	0.625	62.5%
$\frac{3}{4}$	0.75	75%	$\frac{2}{3}$	0.666...	$66\frac{2}{3}\%$
$\frac{2}{5}$	0.4	40%	$\frac{3}{20}$	0.15	15%

Learn these for Paper 1. Also: $\frac{1}{5} = 0.2 = 20\%$, $\frac{1}{3} = 0.333\dots = 33\frac{1}{3}\%$, $\frac{1}{20} = 0.05 = 5\%$, $\frac{1}{100} = 0.01 = 1\%$.

1.2.1 Fraction Basics (N2)

- **Proper fraction:** the numerator is smaller than the denominator, e.g. $\frac{4}{9}$.
- **Improper fraction:** the numerator is equal to or larger than the denominator, e.g. $\frac{17}{5}$.
- **Mixed number:** a whole number with a proper fraction, e.g. $3\frac{2}{5} = \frac{17}{5}$.

- **Equivalent fractions:** equal in value, e.g. $\frac{2}{3} = \frac{10}{15}$ (top and bottom multiplied by 5).
- **Simplest form:** divide numerator and denominator by their HCF, e.g. $\frac{45}{100} = \frac{9}{20}$ (HCF 5).

Mixed to improper: $a\frac{b}{c} = \frac{ac + b}{c}$, e.g. $4\frac{1}{6} = \frac{4 \times 6 + 1}{6} = \frac{25}{6}$.

Improper to mixed: divide and write the remainder over the denominator, e.g. $\frac{29}{12} = 2\frac{5}{12}$.

1.2.2 Calculating Exactly with Fractions (N2, N8)

Adding and subtracting

Rewrite the fractions with a **common denominator** (the LCM of the denominators keeps numbers small), then add or subtract the numerators. The denominator stays the same.

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$$

Worked Example:

Work out $\frac{2}{3} + \frac{4}{5}$.

LCM of 3 and 5 is 15: $\frac{10}{15} + \frac{12}{15} = \frac{22}{15} = 1\frac{7}{15}$

Worked Example:

Work out $4\frac{1}{6} - 1\frac{3}{4}$.

As improper fractions: $\frac{25}{6} - \frac{7}{4} = \frac{50}{12} - \frac{21}{12} = \frac{29}{12} = 2\frac{5}{12}$

Multiplying and dividing

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd} \qquad \frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$$

Change mixed numbers to improper fractions first. To divide, multiply by the **reciprocal** of the second fraction. Cancel common factors before you multiply.

Worked Example:

Work out $3\frac{1}{3} \times 1\frac{2}{5}$.

$$\frac{10}{3} \times \frac{7}{5}: \text{cancel 10 and 5 to give } \frac{2}{3} \times \frac{7}{1} = \frac{14}{3} = 4\frac{2}{3}$$

Worked Example:

Work out $2\frac{1}{4} \div 1\frac{4}{5}$.

$$\frac{9}{4} \div \frac{9}{5} = \frac{9}{4} \times \frac{5}{9} = \frac{5}{4} = 1\frac{1}{4}$$

Worked Example:

Work out $-1\frac{1}{2} \times 2\frac{2}{3}$.

$$-\frac{3}{2} \times \frac{8}{3} = -\frac{24}{6} = -4 \text{ (different signs, so the answer is negative).}$$

Worked Example:

A rectangle is $2\frac{2}{3}$ m long and $4\frac{1}{2}$ m wide. Find its exact area and exact perimeter.

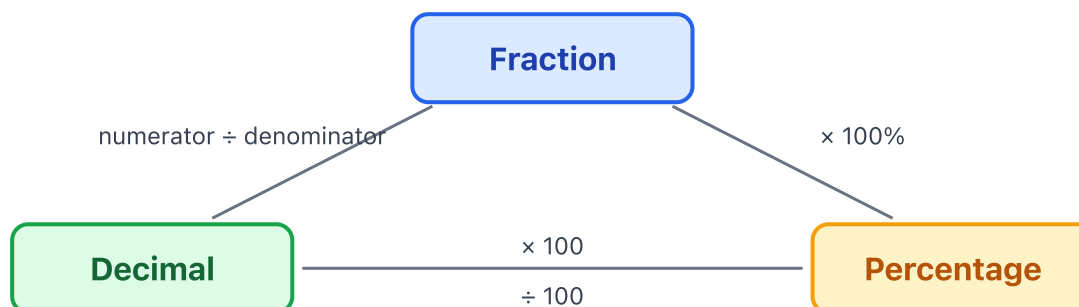
$$\text{Area} = \frac{8}{3} \times \frac{9}{2} = \frac{72}{6} = 12 \text{ m}^2$$

$$\text{Perimeter} = 2 \left(\frac{8}{3} + \frac{9}{2} \right) = 2 \left(\frac{16}{6} + \frac{27}{6} \right) = \frac{43}{3} = 14\frac{1}{3} \text{ m}$$

Exam Tip: "Give your answer as an exact value" or "in its simplest form" means a fully simplified fraction or mixed number, not a rounded decimal. Never add denominators: $\frac{2}{5} +$

$\frac{1}{3} \neq \frac{3}{8}$. For a negative mixed number convert the whole thing: $-2\frac{1}{4} = -\frac{9}{4}$.

1.2.3 Terminating Decimals and Fractions (N10)



Converting between the three forms. Decimal to fraction: write over 10, 100 or 1000 using place value, then simplify.

- **Fraction → decimal:** divide (short division), or find an equivalent fraction over 10, 100 or 1000: $\frac{3}{20} = \frac{15}{100} = 0.15$.
- **Terminating decimal → fraction:** $0.625 = \frac{625}{1000} = \frac{5}{8}$; $4.25 = 4\frac{1}{4} = \frac{17}{4}$.
- **Percentage:** out of 100: $28\% = \frac{28}{100} = \frac{7}{25} = 0.28$.

Worked Example:

Write in order, smallest first: 0.71 , $\frac{7}{10}$, 72% , $\frac{5}{7}$

As decimals: 0.71 , 0.7 , 0.72 , $0.714\dots$

Answer: $\frac{7}{10}$, 0.71 , $\frac{5}{7}$, 72%

Exam Tip: When converting a decimal to a fraction, the last digit tells you the denominator: 0.625 ends in the thousandths column, so start with $\frac{625}{1000}$. Then simplify fully — $\frac{625}{1000}$ alone may not get the final mark.

1.2.4 Recurring Decimals and Fractions (N10)

Higher

A **recurring decimal** repeats a digit or block of digits for ever. Dots are placed over the first and last digit of the repeating block:

$$0.\dot{4} = 0.444\dots \quad 0.\dot{6}\dot{3} = 0.636363\dots \quad 0.8\dot{3} = 0.8333\dots \quad 0.\dot{3}7\dot{0} = 0.370370\dots$$

Which fractions terminate?

Write the fraction in its simplest form. It gives a **terminating** decimal only if the denominator has no prime factors other than 2 and 5; otherwise it **recurs**.

Fraction	Denominator	Decimal
$\frac{9}{40}$	$2^3 \times 5$	0.225 (terminates)
$\frac{5}{12}$	$2^2 \times 3$	0.41666... (recurs)
$\frac{7}{15}$	3×5	0.4666... (recurs)

Fraction → recurring decimal

Use short division, writing extra zeros after the decimal point, until a remainder repeats.

Worked Example:

Write $\frac{2}{11}$ as a recurring decimal.

$2.0000 \div 11$: $20 \div 11 = 1 \text{ r } 9$; $90 \div 11 = 8 \text{ r } 2$; $20 \div 11 = 1 \text{ r } 9$... the remainder 9 has come back, so the digits repeat.

$$\frac{2}{11} = 0.181818\dots = \mathbf{0.\dot{1}\dot{8}}$$

Recurring decimal → fraction

1. Let x equal the recurring decimal, writing out at least two cycles.

2. Multiply by a power of 10 so that one copy has the recurring block just after the decimal point and another copy lines up with it ($\times 10$ for one repeating digit, $\times 100$ for two, $\times 1000$ for three).
3. Subtract so the recurring tails cancel.
4. Solve for x and simplify.

Worked Example:

Prove that $0.\dot{6}\dot{3} = \frac{7}{11}$.

$$\begin{aligned}x &= 0.636363\dots \\100x &= 63.636363\dots \\99x &= 63 \\x &= \frac{63}{99} = \frac{7}{11}\end{aligned}$$

Worked Example:

Write $0.8\dot{3}$ ($= 0.8333\dots$) as a fraction in its simplest form.

The repeating digit starts one place after the point, so use $100x$ and $10x$:

$$\begin{aligned}100x &= 83.333\dots \\10x &= 8.333\dots \\90x &= 75 \\x &= \frac{75}{90} = \frac{5}{6}\end{aligned}$$

Worked Example:

Show that $0.2\dot{5}\dot{4} = \frac{14}{55}$.

$$\begin{aligned}
 1000x &= 254.5454\dots \\
 10x &= 2.5454\dots \\
 990x &= 252 \\
 x &= \frac{252}{990} = \frac{14}{55}
 \end{aligned}$$

Pattern	Example
$0.\dot{a} = \frac{a}{9}$	$0.\dot{4} = \frac{4}{9}$
$0.\dot{a}\dot{b} = \frac{ab}{99}$	$0.\dot{1}\dot{8} = \frac{18}{99} = \frac{2}{11}$
$0.\dot{a}b\dot{c} = \frac{abc}{999}$	$0.\dot{3}7\dot{0} = \frac{370}{999} = \frac{10}{27}$

Exam Tip: Edexcel "Prove algebraically" questions need the full method: define x , write the two multiples, subtract, and simplify. Quoting the pattern above gains no marks for a proof. Show enough repeating digits that the tails clearly cancel.

1.2.5 Fractions and Percentages as Operators (N12)

"Of" means multiply. When a fraction or percentage acts on an amount it is an **operator** — a multiplier.

Fraction of an amount: $\frac{3}{8}$ of 56 = $56 \div 8 \times 3 = 21$

Percentage of an amount: $r\%$ of $A = \frac{r}{100} \times A$, e.g. 22.5% of 160 = $0.225 \times 160 = 36$

Increase by $r\%$: multiply by $(1 + \frac{r}{100})$ Decrease by $r\%$: multiply by $(1 - \frac{r}{100})$

Change	Multiplier
Find 6%	0.06

Increase by 7%	1.07
Increase by 20% (e.g. adding VAT)	1.2
Decrease by 17.5%	0.825
Decrease by $\frac{2}{5}$	$\frac{3}{5}$ (or 0.6)

Worked Example:

Without a calculator, work out 45% of £72.

10% = £7.20, so 40% = £28.80. 5% = £3.60. 45% = 28.80 + 3.60 = **£32.40**.

Worked Example:

A flat worth £240 000 increases in value by 5% and then decreases by 5%. Find its final value.

$$240\,000 \times 1.05 \times 0.95 = \textbf{£239\,400}$$

The 5% decrease is taken from the larger value £252 000, so the flat does not return to £240 000. Overall multiplier $1.05 \times 0.95 = 0.9975$, a 0.25% decrease.

Worked Example:

A pair of trainers costs £85. A shop takes 20% off, then a further 10% off the sale price. Find the final price.

$$85 \times 0.8 \times 0.9 = \textbf{£61.20} \text{ — not the same as 30\% off, which gives } 85 \times 0.7 = \textbf{£59.50}.$$

Exam Tip: Successive percentage changes are multiplied, never added. A single multiplier is the quickest method on Papers 2 and 3. Percentage change, reverse percentages and compound interest belong to the ratio and proportion topics (R9, R16).

1.2.6 Fractions in Ratio Problems (N11)

A ratio compares **part with part**; a fraction compares a **part with the whole**.



Ravi : Sita = 3 : 4. Ravi gets $\frac{3}{7}$ of the total and Sita gets $\frac{4}{7}$; Ravi's share is $\frac{3}{4}$ of Sita's.

- Ratio $a : b \rightarrow$ the first share is $\frac{a}{a+b}$ of the whole.
- If $\frac{5}{9}$ of a club are girls, then $\frac{4}{9}$ are boys and girls : boys = 5 : 4.
- "A gets $\frac{3}{4}$ of what B gets" means $A : B = 3 : 4$.

Worked Example:

Ravi and Sita share £154. Ravi gets $\frac{3}{4}$ of the amount Sita gets. Work out how much each receives.

Ravi : Sita = 3 : 4, a total of 7 parts. One part = $154 \div 7 = 22$.

Ravi = $3 \times 22 = \text{£}66$, Sita = $4 \times 22 = \text{£}88$.

Worked Example:

Adults and children at a pool are in the ratio 4 : 5. $\frac{1}{2}$ of the adults and $\frac{2}{5}$ of the children are swimming. What fraction of all the people are swimming?

Adults are $\frac{4}{9}$ of the people: $\frac{1}{2} \times \frac{4}{9} = \frac{2}{9}$. Children are $\frac{5}{9} \times \frac{2}{5} = \frac{2}{9}$.

Total swimming = $\frac{2}{9} + \frac{2}{9} = \frac{4}{9}$.

1.2.7 Multi-step Fraction Problems (N8, N12)

When a fraction is taken of "the rest" or "the remainder", convert each part into a fraction of the *original* amount before you work with money or quantities.

Worked Example:

Jess spends $\frac{1}{3}$ of her money on travel and $\frac{3}{4}$ of the remainder on food. She has £45 left. How much did she start with?

After travel, $\frac{2}{3}$ remains. She keeps $\frac{1}{4}$ of that: $\frac{2}{3} \times \frac{1}{4} = \frac{1}{6}$ of the original.

$\frac{1}{6}$ of the total = £45 $\Rightarrow$ total = **£270**.

Worked Example:

A water butt is $\frac{5}{6}$ full. After 35 litres are used it is $\frac{1}{4}$ full. Work out its capacity.

$\frac{5}{6} - \frac{1}{4} = \frac{10}{12} - \frac{3}{12} = \frac{7}{12}$ of the butt is 35 litres.

$\frac{1}{12} = 5$ litres, so capacity = **60 litres**.

Exam Tip: A bar model makes "remainder" problems much clearer: shade the first fraction, split what is left into the second fraction's parts, and label the final piece with the known amount. Check by working forwards from your answer.

Summary

Skill	Key method
Add / subtract fractions (N2, N8)	Common denominator, then combine numerators
Multiply / divide fractions (N2, N8)	Improper fractions; multiply across; divide = multiply by the reciprocal

Terminating decimals (N10)	Fraction $\rightarrow$ decimal: divide; decimal $\rightarrow$ fraction: place value then simplify
Operators (N12)	"Of" means $\times$; increase by $r\%$: $\times \left(1 + \frac{r}{100}\right)$
Ratio and fractions (N11)	Share $\div$ total number of parts
Recurring decimals (N10) Higher	x , $10^n x$, subtract, solve, simplify